

Aerospace Engineering Graduate Programs — Application Guide

For Olivia — applying with a theoretical physics master's from NBI, targeting Fall 2027 admission.

How AE graduate admissions work

AE admissions in the US are broadly similar to physics PhD programs — you apply to the department, not to a specific advisor, though identifying faculty you want to work with is important and expected. A few differences:

- Master's vs. PhD: In the US, most funded positions are PhD programs. Standalone master's programs often require you to pay tuition or are unfunded. A funded PhD is always preferable if you're committed to research. Some students enter as MS and transition to PhD; others enter directly into PhD programs. Clarify with each program.
 - Engineering GRE: Some programs still request a general GRE. An engineering-specific GRE is rare now. Check each program's current policy.
 - Letters of recommendation: Three, ideally from research supervisors. Oders and Holger Bech Nielsen at NBI are strong. Your CERN supervisor counts.
 - Statement of Purpose: More important than the GRE. Should articulate what sub-area you want to work in and why — and name specific faculty. Be honest about your physics background and direct about where you want to take it.
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Your profile — how to present it

You are not a typical AE applicant. That is an asset if framed correctly.

What works for you:

- NBI master's in theoretical physics — signals mathematical maturity and serious research capability
- CERN research experience — institutional name recognition, concrete research credential
- Bachelor's thesis on string theory and black holes — demonstrates ability to engage with hard theoretical problems
- Strong mathematical background — directly applicable to electric propulsion, astrodynamics, space environment

What to address:

- No prior AE or engineering coursework — acknowledge it directly in the Statement of Purpose, then pivot to why your background is relevant to the specific sub-area you're targeting. "I come from a plasma physics background, which I believe is directly

applicable to Hall thruster research" is more effective than silence on the gap.

- Be specific about sub-area — "aerospace engineering" is too broad. "Electric propulsion, specifically plasma instabilities in Hall thrusters" or "astrodynamics for cislunar mission design" tells the committee where to route your application and who to show it to.
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The master's as a bridge — why this path makes sense

Coming from a theoretical physics background, a master's in aerospace engineering is not a detour. It is the correct path. It fills the engineering fundamentals gap (aerodynamics, thermodynamics, structures, dynamics), produces a credential that opens PhD doors, and gives you time to find out which sub-field actually pulls before committing to a five-year dissertation.

The academic challenges are real but manageable. The math won't be the problem — a string theory master's leaves you ahead of most AE students in mathematical maturity. The harder adjustment is physical intuition in fluid mechanics and aerodynamics, engineering thermodynamics, and lab culture (the shift from deriving equations to making experiments produce useful data). The first semester of an AE master's is the crunch point. By the second semester you find your footing.

Funding reality for AE master's programs

This is the most important thing to understand before building a list:

Most US AE master's programs are not funded. This is a critical difference from physics PhD programs, where essentially everyone admitted gets a stipend. An unfunded AE master's at MIT or Stanford costs \$60,000/year in tuition alone.

Options that solve this:

1. Enroll in a PhD program and earn the MS along the way. Many AE PhD programs award a master's after the first two years. The PhD is funded; the MS comes with it. This requires committing to a PhD before you're certain — but financially it's the cleanest US option.
 2. Find a faculty member who funds you as a research assistant. Some professors take funded master's students on grants. Not guaranteed, but worth asking explicitly in faculty outreach emails. The language: "I'm interested in a research master's — do you have grant funding that could support a student working in this area?"
 3. European master's programs (strongly recommended). TU Delft and KTH treat you as an employee or charge near-zero EU tuition. The degree is excellent. The cost is a fraction of US programs. This is not a consolation — it is often the smarter choice.
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Master's programs by fit for a physics background

TU Delft — MSc Aerospace Engineering (Netherlands) ★

The standout option for someone in Olivia's position. Two-year research master's, EU tuition (~€2,200/year), genuinely world-class faculty, strong pipeline into US PhD programs. Olivia would spend two years in the Netherlands building the engineering foundation properly — aerodynamics, propulsion, structures, GNC — with research embedded from the start. She'd graduate at 27 with a complete AE profile, then apply to US PhD programs with full clarity about sub-area and direction.

A Delft MSc followed by a US AE PhD application is a well-worn, credible path. Many top US PhD programs know Delft graduates well.

Sub-specialisations available: Aerospace Structures and Materials, Aerodynamics, Space Exploration, Flight Performance and Propulsion, Control and Simulation. The Space Exploration track maps directly to her interests.

KTH — MSc Aerospace Engineering (Sweden)

Free for EU citizens. Two-year program in Stockholm. Somewhat less strong than Delft in AE specifically but excellent by any normal standard, and the cost advantage is significant. A good option if proximity to Scandinavia matters or if the Stockholm environment appeals.

MIT AeroAstro SM (USA)

Thesis-based, typically 1.5–2 years. Rarely funded unless a faculty member takes you on as an RA. The most prestigious AE master's in the US. Getting in from a physics background is possible but not straightforward — the application should emphasise mathematical depth and plasma physics relevance to propulsion or space research. If funded, exceptional. If unfunded, \$60,000/year in tuition makes it very hard to justify over Delft.

Georgia Tech MSAE (USA)

Large, structured, strong foundational coverage across all AE sub-fields. More accessible than MIT or Stanford. Good for someone who needs broad exposure before narrowing. Rarely funded as a standalone MS. Atlanta is affordable; the research culture is strong.

Michigan MS Aerospace Engineering (USA)

Access to PEPL (electric propulsion lab) and space weather group — directly relevant for the sub-areas where a physics background lands well. Sometimes funded via RA if a faculty member in those groups has space. Ann Arbor is excellent and affordable.

CU Boulder MS Aerospace Engineering Sciences (USA)

Space-focused culture, strong astrodynamics group, LASP next door. A natural environment

for someone interested in space propulsion or space environment. Sometimes funded. Boulder is one of the best places to live in the US.

The master's pathway timeline

Option	Timeline	Cost	Outcome
TU Delft MSc → US PhD	2027–2029 MS; 2029–2034 PhD	Near-zero MS cost; funded PhD	Enters PhD at 27 with full engineering foundation
KTH MSc → US PhD	Same	Free MS; funded PhD	Same
US MS (unfunded) → US PhD direct (MS embedded)	2027–2029 MS; 2029–2034 PhD	\$100–120k MS cost	Same outcome, much higher cost
US PhD direct (MS embedded)	2027–2032+	Funded throughout	Skips standalone MS; riskier without engineering base

The TU Delft path is the most financially rational and academically well-matched option for this specific profile. It is also the most honest: two years to build the foundation properly, then enter a PhD knowing exactly what you're doing and why.

Timeline for Fall 2027 admission

When	What
Summer 2026 (now)	Research programs and faculty; identify 2–3 target faculty per school
August–October 2026	Email faculty; introduce yourself and your background
September 2026	Ask recommenders (Obers, Holger Bech Nielsen, CERN supervisor)
October–November 2026	Draft Statement of Purpose
December 2026–January 2027	Most US AE deadlines fall here; submit early
February–March 2027	Interview invitations and visit days

Contacting faculty

Do this — same advice as for physics programs, same timing.

For AE, the research hook is even more important because your background is non-standard. Faculty want to know you've read their work and understand what they actually do. A specific email from someone with a plasma physics background asking a genuine question about Hall thruster instabilities will get read. A generic email from a physicist saying they're interested in "aerospace" will not.

Template: same structure as the physics email templates. The key adaptation — be explicit about your physics background and why it connects to their specific research rather than the field generally.

The background gap — practical steps

If you want to strengthen the engineering side before applications or during a master's:

- Online coursework: MIT OpenCourseWare has full aerodynamics, thermodynamics, and orbital mechanics courses. Working through one or two before or during applications shows initiative and fills vocabulary gaps.
 - arXiv and AIAA papers: Read papers from the groups you're targeting. Understanding the language and open problems before you email faculty makes the email substantively better.
 - Talk to Obers: He may know people in the AE community through interdisciplinary connections, particularly in space plasma physics. Ask.
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International programs — key differences

TU Delft (Netherlands): Apply through the TU Delft Graduate School. Programs in English. EU tuition (~€2,200/year). Competitive but less so than top US programs. Excellent option.

ISAE-SUPAERO (France): Apply directly. Programs increasingly in English. EU tuition. Research master's available. Strong industry ties.

KTH (Sweden): EU tuition (free for EU citizens). Apply through the KTH admissions portal. Programs in English. Excellent quality of life.

Imperial College (UK): Post-Brexit, Danish students pay international fees (~£32,000/year for

engineering). Requires scholarships to be viable. The Schrödinger Scholarship and other external awards are routes.

Money — what to expect and what to assume

The first distinction: stipend vs. scholarship

A stipend comes with a funded PhD position — it is the regular payment that accompanies the research appointment. A scholarship is a separate competitive award you apply for independently. These are not the same thing, and you should not assume one is the other.

US PhD programs

Most funded PhD programs in AE cover tuition and pay a stipend. These are research assistant (RA) or teaching assistant (TA) appointments — employment, not charity. If you are admitted to a funded PhD position, the stipend comes with it automatically.

The critical point: standalone master's programs in US AE are often unfunded. You pay tuition. Some students enter as MS and hope to transition to a funded PhD — this is a real path but carries financial risk. Apply to PhD programs if you want funding.

Approximate stipend ranges for funded PhD positions (not guaranteed until you have an offer):

Program	Stipend/yr	City cost
MIT AeroAstro	~\$40–44k	High (Boston)
Princeton MAE	~\$40–44k	Moderate (NJ)
Caltech GALCIT	~\$38–42k	High (LA)
CU Boulder	~\$28–33k	Moderate
Georgia Tech	~\$28–32k	Moderate (Atlanta)
UT Austin	~\$26–30k	Moderate-high
Michigan AE	~\$25–30k	Low (Ann Arbor)
Purdue AAE	~\$22–26k	Very low (W. Lafayette)

European programs

At TU Delft and KTH, PhD students are legally employees — the salary is attached to a specific funded position advertised by a supervisor, not to the institution generally. You cannot simply apply to the university and expect to be paid. You need to be selected for a specific funded position, or have a supervisor with grant money who agrees to take you.

If you find a funded position: TU Delft pays ~€2,500–2,800/month; KTH pays ~30,000–35,000 SEK/month. Both cover you as an employee with benefits. But finding the position is the work.

ISAE-SUPAERO research master's funding varies by program and year — do not assume funding; check each program explicitly.

Imperial London: funded positions exist but are competitive. Unfunded, London is very expensive on a student budget.

Tuition

For funded PhD positions, tuition is not your cost — it is either explicitly waived as part of the funding package (US) or covered by the institution as part of your employment contract (Netherlands, Sweden). You pay nothing.

Tuition becomes real money in two situations:

Unfunded US master's:

Program	Unfunded MS tuition/yr
MIT	~\$60,000
Caltech	~\$60,000
Princeton	~\$58,000
Michigan	~\$23-48k (in/out of state)
CU Boulder	~\$15-30k
Georgia Tech	~\$14-32k
UT Austin	~\$12-25k
Purdue	~\$10-29k

This is why applying to funded PhD programs matters. An unfunded MS at MIT costs roughly what a house costs in West Lafayette.

Imperial College London: ~£32-35k/year in tuition as an international student (post-Brexit), regardless of whether a stipend is attached. A funded position covers it; without one, it is prohibitive.

EU institutions: TU Delft charges ~€2,200/year for EU students; KTH is free for EU citizens. For funded PhD positions at either, tuition is covered by the institution. Effectively zero either way.

The Danish comparison

Danish PhD students are employees and earn ~390,000-420,000 DKK/year (~\$56-61k) with full benefits. Every option above is a nominal pay cut from that. The trade-off is the institution, the supervisor, and where the degree takes her.

See also: [landscape.md](#) for sub-field descriptions, [programs.md](#) for program details